# EchoPulse: Pioneering Precision in Pulmonary Diagnostics through Digital Auscultation Using Random Forest and SVM Models

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## ABSTRACT

Cardiopulmonary diseases impact 576.3 million individuals globally, having a disproportionate impact on low and middle-income countries (LMICs) due to increased exposure to risk factors and inaccessible diagnostic methods (World Health Organization, 2019). Inaccessibility in these regions stems from inadequate monetary funds and sparsity of locations offering diagnostic testing, in turn leading to late-stage diagnoses and poor patient outcomes. This paper introduces a novel all-in-one device, *EchoPulse*, meticulously designed with a focus on affordability, ease-of-use and accuracy to diagnose heart failure, chronic obstructive pulmonary disease (COPD) and asthma. *EchoPulse* leverages a multitude of advanced technologies including machine learning models, a Random Forest classifier and an SVM model, with hardware components such as a Raspberry Pi to diagnose cardiopulmonary diseases with a 92% accuracy rate. The paper further delves into the benefits of launching a pilot project in Papua New Guinea to demonstrate the efficacy and impact of *EchoPulse* in improving healthcare outcomes in resource-constrained environments and using those learnings to prepare for a widespread launch in other LMICs.

**Keywords:** Cardiopulmonary diseases, heart failure, COPD, asthma, diagnostic tool, Random Forest, SVM
**Note:** All currencies in Canadian Dollars unless otherwise noted

## 1. INTRODUCTION

Cardiopulmonary diseases encompass a spectrum of disorders affecting both the heart and lungs, posing tremendous health challenges globally. Among these diseases, the most prominent are heart failure, COPD, and asthma. Heart failure occurs when the heart is unable to pump adequate blood supply to meet the body's needs; COPD is characterized by progressive airflow limitation in the lungs that is not inherently reversible; and asthma is a chronic inflammatory disorder of the airways (Mayo Clinic, 2023)

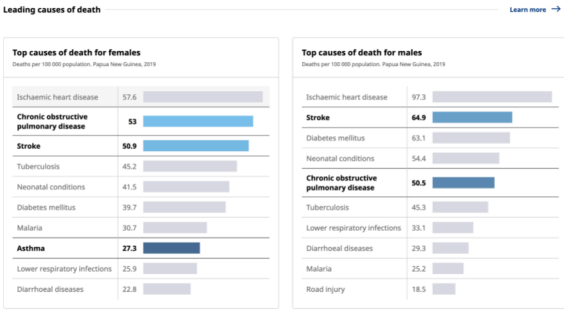


Figure 2: All lung and heart heart disease are highly prevalent in Papua New Guinea but with high number of COPD and Asthma

*Figure 1: Cardiopulmonary diseases are among the top causes of death in Papua New Guinea for both Females and Males*

According to the World Health Organization, cardiopulmonary diseases collectively impact over one million individuals in Papua New Guinea. (World Health Organization, 2019). Figure 1 shows heart failure, COPD, and asthma are the leading causes of death among both females and males in the country. The primary risk factors for increased chances of cardiopulmonary diseases include unhealthy dietary habits, inactive lifestyles, overconsumption of tobacco and alcohol, and prolonged exposure to air pollution.

In conjunction with several risk factors, current diagnostic methods for Cardiopulmonary disease in the market like X-rays and CT scans are very expensive, due to the high cost of machinery

around $500,000 (*Micro Photonics*, 2015). As a result, these diagnostic methods are highly inaccessible to the majority of the population in many developing countries. The World Health Organization reports two-thirds of the world population does not have access to basic radiology services. In Papua New Guinea, 80% of the population, roughly 7.2 million people, must undertake a multi-day walk to access basic X-ray services. This inaccessibility results in over 600,000 cases of heart failure, COPD and asthma remaining undetected and undiagnosed (Silverstein, 2016), in turn making it difficult to focus on prevention efforts to halt the severity of the disease progression.

As the burden of cardiopulmonary diseases continues to escalate, particularly in regions like Papua New Guinea where access to diagnostic services remains limited, there is an urgent need for innovative, affordable, and accessible diagnostic solutions. *EchoPulse* emerges as a promising response to this critical need. *Echopulse* has leveraged advanced machine-learning algorithms to revolutionize the diagnosis of cardiopulmonary conditions. By utilizing cutting-edge advanced technology with a focus on affordability and accessibility, *EchoPulse* has the potential to dramatically improve healthcare outcomes in resource-constrained areas.

## 2. BACKGROUND

Current diagnostic methods for cardiopulmonary diseases such as heart failure, COPD, and asthma pose significant challenges due to their high cost, inaccessibility and limitations in early detection. For heart failure, diagnostic tools include chest X-rays, electrocardiograms (ECG), and CT scans of the heart alongside various blood tests. However, these methods are often expensive and inaccessible, particularly in resource-constrained settings. ECGs can cost up to $600 (Ontario, OR *EKG Cost Average*, 2024).

Similarly, diagnosing COPD typically involves spirometry tests, chest X-rays, and CT scans. Spirometry, the most common test for COPD, only detects symptoms in later stages, leading to delayed treatment and increased mortality risk. According to studies, delayed treatment for chronic diseases like COPD can nearly double the risk of mortality (Seemungal et al., 2009). Chest X-rays and CT scans are accurate for diagnoses but costly and inaccessible. CT scans, which provide detailed images of the lungs range from $500 to $1,000 (*Private CT Clinic and Procedures*, 2023).

For asthma diagnosis, spirometry and peak flow tests are commonly employed to measure lung function. However, these tests may not detect symptoms until they have progressed significantly, leading to delayed treatment. Additional imaging tests such as chest X-rays are required to confirm diagnoses, however, X-rays of the chest can cost over $1,000 (*Ontario X-Ray Cost Average*, 2024).

Overall, the current diagnostic landscape for cardiopulmonary diseases is characterized by expense, inaccessibility, and limitations in early detection. This highlights the urgent need for innovative and accessible diagnostic solutions.

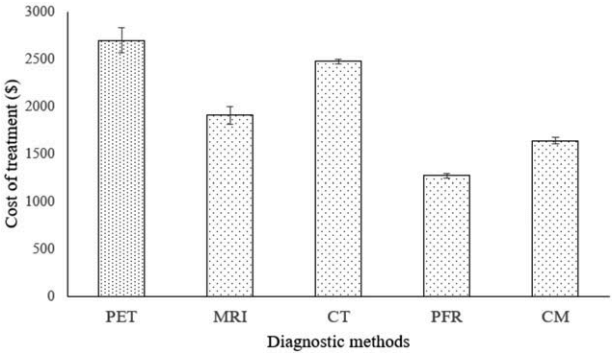


*Figure 2: Price in USD of Common Diagnostic Methods*

## 3. METHODOLOGY OF DATA COLLECTION

The data was collected by the Jordan University of Science and Technology's Faculty of Computer and Information Technology. The study collected high-quality lung and heart sounds from both healthy and unhealthy patients using a digital stethoscope, which has been pivotal in recording and identifying accurate sounds associated with

cardiopulmonary conditions. The dataset is comprised of audio recordings of lung and heart sounds for several illnesses including asthma, heart failure, pneumonia, bronchitis, pleural effusion, lung fibrosis, and chronic obstructive pulmonary disease (COPD), alongside recordings of healthy patients.

The study involved 112 individuals, of whom 35 were healthy, and 77 had been diagnosed with lung conditions. These subjects ranged in age from 21 to 90 years, with an average age of 50.5 years, encompassing 43 females and 69 males. The data files were systematically labelled to reflect the filter type, diagnosis, sound type, chest location, individual's age, and gender, enabling precise data identification, categorization, and retrieval.

Each heart and lung sound recording was processed through three different frequency filters to highlight specific sound ranges critical for diagnosing conditions. The filters included Bell mode, Diaphragm mode, and Extended mode, each mode emphasizing different frequency ranges to capture a comprehensive spectrum of bodily sounds. This tripartite filtration approach facilitated a nuanced analysis of the lung sounds, aiding in the accurate detection and categorization of the various pulmonary diseases.

To enhance the dataset and improve the robustness of the model, data augmentation techniques were applied using *Python* to increase the diversity of the audio samples. This process involved artificially expanding the dataset by generating modified versions of the original recordings, thereby simulating a wider range of auditory scenarios that could be encountered in clinical settings. Specifically, augmentation was achieved by adding background noise to the recordings and altering the pitch of the lung sounds. Adding noise helped simulate different environmental conditions, while pitch alteration mimicked variations in patient lung sounds due to age, size, or severity of the condition. These manipulations were carefully calibrated to ensure that the augmented data remained both realistic and medically relevant. Through these augmentation techniques, the dataset was improved, offering a broader spectrum of audio samples. This expansion played a crucial role in training the model, enhancing its ability to generalize from the training data and improving its accuracy in recognizing and classifying a wide array of lung sounds under various acoustic conditions.

## 4. SOFTWARE PROTOTYPE

*EchoPulse* meticulously processes audio data from lung sounds to identify respiratory conditions. Initially, it extracts audio features using Mel-frequency cepstral coefficients (MFCCs), a method pivotal for capturing sound patterns. This is supplemented by augmenting the audio data through various techniques like adding noise and altering the pitch, enhancing the model's exposure to varied data representations.

The Random Forest (RF) classifier is a predictive model that combines predictions of multiple decision trees, accounting for different audio signals, like coughs, crackles and murmurs, to provide an accurate prediction. The system optimizes the RF model by adjusting hyperparameters like the number of trees, tree depth, and the minimum number of samples for splitting nodes, using a comprehensive grid search to identify the most effective settings.

A support Vector Machines (SVM) is stacked on top of these models and plays a pivotal role in EchoPulse's disease classification system by identifying an optimal hyperplane in the feature space. Given the intricate nature of disease categorization through audio, a linear boundary proves inadequate, prompting SVM to employ the kernel trick. This technique transforms data into a higher-dimensional space using functions such as the radial basis function (RBF) kernel, $K(x, x') = \exp(-\gamma \, || x - x' ||^2)$. This transformation enables SVM to capture non-linear relationships

among audio features, facilitating the construction of complex decision boundaries essential for accurate disease classification. Support vectors, critical to SVM operation, are pivotal audio samples closest to the hyperplane, defining the margin between disease categories. During the classification of new audio recordings, EchoPulse's SVM model assesses disease classification by evaluating where its features lie relative to this hyperplane, ensuring precise and efficient diagnosis.

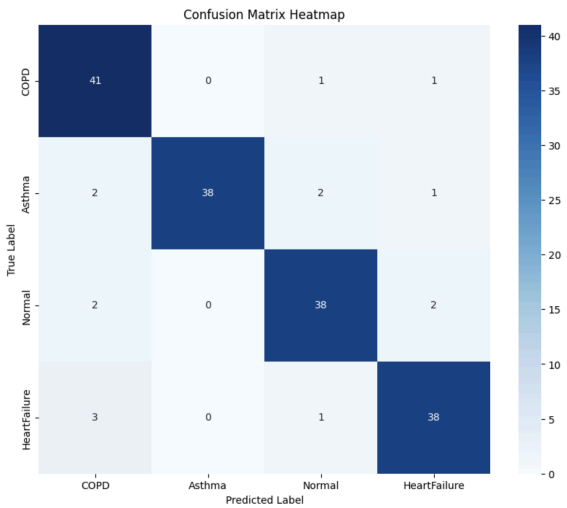


*Figure 3: Confusion Matrix showcasing 92% accuracy*

*EchoPulse* evaluates both models independently, using metrics like accuracy at 92% and confusion matrices to gauge the model's performance (refer to Figure 3). It combines their predictive strengths, averaging their outputs to yield an accurate and reliable diagnosis, taking advantage of the SVM's and the RF's detailed feature analysis. This dual-model strategy embodies *EchoPulse's* comprehensive and nuanced approach to audio-based disease classification.

## 5. DESIGN PROTOTYPE

The Raspberry Pi 3B is the main focus of this prototype of the digital stethoscope (Refer to Figure 4). It acts as the central processing unit (CPU), playing a role to process the audio input recorded by the USB collar microphone capturing lung sounds (C). The tool is able to communicate with the user through a 3.5-inch touchscreen display (E) when a microspeaker is unavailable, providing a simple interface.

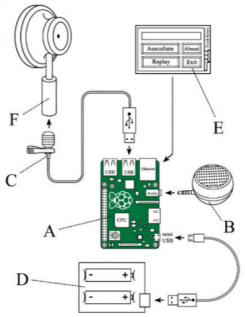


Figure 2: (A) Raspberry Pi 3B. (B) Microspeaker (Optional) (C) USB collar microphone. (D) Uninterruptible power supply battery expansion board with two rechargeable batteries (18650). (E) 3.5-inch touch screen. (F) Chest piece from a conventional stethoscope. CPU: central processing unit.

*Figure 4: Final Hardware Mock-up Design*

Through the CPU, the audio data is then filtered and amplified, where it is then displayed on the interface. After this process in the *EchoPulse* device, the in-built model runs the classification algorithms. These results are then seen visually on the touch screen, so the user can easily analyze the data.

Because of the power battery expansion board (D) with two rechargeable batteries (18650), the device is able to operate without an external power source. These rechargeable batteries may be replaced with a portable charging bank. This feature is crucial in healthcare environments as flexibility is one of the most important aspects as it often aids with increased accessibility. The chest component of a traditional stethoscope (F) is also seen in this design, playing the role of the contact point to record physiological sounds. This part was added to the electronic system to maintain the well-known use case for medical professionals, which was originally used for acoustic stethoscopes.

## 6. MATERIALS AND PROCEDURE

→ Chest Piece + USB collar microphone
→ Raspberry pi 3B
→ 3.5 inch display screen

1. Attach the USB collar microphone (C) to one of the USB ports on the Raspberry

Pi 3B (A). Ensure the connection is secure as this microphone is essential for capturing heart and lung sounds.
2. Connect either batteries or a portable charger to the Raspberry Pi 3B (A)
3. Once the system has booted, open the *EchoPulse* application on the touch screen. The software is ready to run the diagnostic tests.
4. Use the stethoscope in the same manner as a conventional one—place the chest piece (F) on the patient's chest wall at the required auscultation points.
5. After capturing the lung sounds, the *EchoPulse* software will process the audio data. You may need to wait a few minutes as the system filters, augments, and analyzes the sounds using its SVM and Random Forest models.
6. The screen (E) will display the diagnostic outcome once the analysis is complete.

## 7. PATIENT JOURNEY

The *EchoPulse* digital stethoscope design is aimed to be as flexible and mobile as possible, adapting to environments at NGOs, hospitals, and simply shared use within a neighbourhood.

A patient's journey in using *EchoPulse's* design starts at the point of care, where professionals at NGOs or hospitals use this as part of the usual physical examination. As a result of its mobility, the digital stethoscope can easily be carried to the patient's side, reducing unnecessary movement within the environment. After recording, within a matter of minutes, the device processes the lung sounds and presents the data on the touch screen for decision-making.

In settings distant from the city, the *EchoPulse* device can act as an essential health monitoring tool. Only one device is required for an entire community and can be worked by a trained or designated operator. Similar to the traditional stethoscope, patients are able to check their own lung sounds. After a short wait for analysis, the stethoscope itself displays the results directly to the user, allowing for use without the need for a specialist and under significantly less time for potential diagnosis through traditional means.

Because of its simplicity and quick diagnostics, *EchoPulse's* design shows a growing movement in the patient care industry, especially for places without reliable healthcare. This device both enhances the efficiency of diagnostics, while also improving the experience for users by providing instant analysis.

## 8. COST BREAKDOWN AND ANALYSIS

The overall Bill of Materials (BOM) cost, including manufacturing expenses, adds up to $98.71 (refer to Figure 5). The majority of this cost comprises the Raspberry Pi 3B, the USB collar microphone, and the LCD display screen.

One of the primary benefits of this tool is its affordability, serving as a much cheaper alternative to traditional diagnostic technology. As a result of this, it can reach significantly more people, expanding its impact to more individuals who need cardiopulmonary testing. This design has the potential to affect the lives of numerous people with limited access to medical equipment.

The cost of this tool adds to the importance of guaranteeing access to early detection of cardiopulmonary diseases to a broad population. Early detection is a crucial aspect of managing diseases and improving patient outcomes. Healthcare users can now quickly include this tool for regular screenings because of its lower cost, consequently identifying diseases and enhancing the lives of patients at risk of cardiopulmonary diseases.

In addition, the affordability of this design has benefits for society as a whole. By being accessible to a larger population, it also gains access to a deeper understanding of the frequency of cardiopulmonary diseases. It is able

to collect data for research purposes, thus fueling new innovations in this field and potentially influencing health policies.

| Components | Cost (CAD) |
|---|---|
| Total Cost | $98.71 |
| Raspberry Pi 3B | $48.95 |
| 3.5 Inch Display Screen | $24.81 |
| USB Collar microphone | $19.95 |
| Chest Piece | $5.00 |

*Figure 5: Cost breakdown of the materials required for the prototype.*

## 9. CONCLUSIONS

*EchoPulse's* digital stethoscope design is an extraordinary innovation in the field of medical diagnostics. Because of its advanced hardware and software, this device has a 92% accuracy rate when detecting prominent heart and lung diseases. This system was designed with an emphasis on low cost, efficient functionality, and mobility in worldwide healthcare.

This device is designed to make the patient's experience user-friendly. This device uses a simple, yet effective, process that provides patients with information about their health concerns instantly. This aspect has been important in gaining a better understanding of the diagnosis process, making it less intimidating for patients, especially in developing areas.

In the future, the growth of the *EchoPulse* dataset will cover more respiratory and cardiac diseases, including lung fibrosis, pneumonia, bronchiolitis obliterans, and pleural effusion. This expansion is expected to improve the device's accuracy, as well as its usage rate in the medical field.

The immediate next step for *EchoPulse* is the release of this device in Papua New Guinea, given the high prevalence of lung and heart diseases. Collaborating with NGOs, like KTF and the PNG Tribal Foundation, can help the brand establish credibility in local markets and devise a targeted launch strategy to tap the most saturated markets first to maximize the device's intended health impact. NGO partners can also help manage potential language barriers which inadvertently could impose accessibility issues if not tackled head-on.

In Papua New Guinea, around one million individuals are affected with lung or heart diseases. With an expected usage of 15%, *EchoPulse* can diagnose 150,000 people.

The potential for this device in Papua New Guinea is highly attractive, but it also has the potential to have a significant impact worldwide. The global population of people affected with lung or heart diseases is 576.3 million, and still assuming a usage rate of 15%, this device can diagnose around 86.445 million people around the world.

*EchoPulse's* drive for making high-quality healthcare more accessible and efficient through this design is expected to change the lives of millions, creating a healthier population worldwide.

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